Chapter 1

Reaction Rates & Charaged Particle Induced Reactions

1.1 Reaction Rates

The reaction Q-value determines the amount of energy released or absorbed during a nuclear reaction. However, in stellar environments the energy released by an individual reaction is only one part of the problem. For many astrophysical applications our primary interest lies with how many nuclear actions occurring within a given volume during a given interval of time. This quantity is known as the **reaction rate**. The reaction rate plays a central role in the study of stellar energy generation nucleosynthesis and the evolution of stars. Reaction rates connect to the main objectives of our course[1]

1. Understanding the synthesis of elements in the Universe
2. Understanding the energy produced by stars

The cross-section is defined as the ratio

$$\boxed{\sigma = \frac{\text{Number of interactions per unit time}}{(\text{Incident particles per unit area per unit time})\,(\text{Number of target nuclei})}}$$

The unit of cross section is area, usually expressed in cm^2 or more conveniently in barns, millibarns, microbarns, and nanobarns. When discussing laboratory measurements we often refer to the *target and projectile*. However inside a stellar plasma there is no distinction between target and projectile because all particles are moving simultaneously. Therefore, the stellar problem must be formulated using the relative motion of the interacting particles. Since our main interest in astrophysics is not only the energy released in a single reaction but also the frequency with which reactions occur, we introduce another concept called *reaction rate*

1.1.1 Reaction Rate Nomenclature

The reaction rate is defined as the number of reactions occurring per unit volume per unit time. It is the most important quantities in Nuclear Astrophysics course because it determines

- the abundance evolution of elements
- the energy production inside stars
- the rate of stellar nucleosynthesis

The reaction rate depends mainly on two quantities: First the reaction crosss section and then velocity distribution of particles in the stellar plasma. At finite temperature, particles possess thermal kinetic energy and move in different directions. Consequently, collisions occur continuously, and some of these collisions lead to nuclear reactions. Consider generic nuclear reaction

$$\boxed{1 + 2 \to 3 + 4}$$

In laboratory terminology

[1] Main objectives of this course is to understand the the origin of the elements in our universe but not as a todler

- Particle 1 is the projectile
- Particle 2 is the target nucleus
- Particles 3 and 4 are the reaction products

Particle 4 may be a proton, neutron, alpha particle, or gamma ray, while particle 3 is usually the residual nucleus. The reaction cross-section depends on the relative energy of the interacting particles and may therefore be written as $\sigma(E)$, or equivalently $\sigma(v)$. The reaction rate is defined as

$$r_{12} = \frac{N_R}{tV} \qquad \text{reaction rate} = \text{reactions per unit volume per unit time}$$

where

- N_R is the number of reactions
- t is the time interval
- V is the interaction volume

The beam current density is defined as

$$j_b = \frac{N_b}{tA}$$

where N_b is the number of beam particles and A is the beam area. Using the definition of cross-section

$$\frac{N_R}{t} = \sigma j_b N_t$$

where N_t is the number of target nuclei. Substituting the expression for j_b

$$\frac{N_R}{t} = \sigma \frac{N_b}{tA} N_t$$

Dividing by the volume V

$$r_{12} = \frac{N_R}{tV} = \sigma \frac{N_t}{V} \frac{N_b}{V} v$$

Defining the number densities

$$N_1 = \frac{N_t}{V} \qquad N_2 = \frac{N_b}{V}$$

the reaction rate becomes

$$\boxed{r_{12} = N_1 N_2 v \sigma(v)}$$

where N_1 and N_2 are the number densities of the interacting particles.

1.1.2 Velocity Distribution

At thermodynamic equilibrium, all particles do not move with the same velocity. Instead the relative velocities are described by a probability distribution $P(v)$. The quantity $P(v)\,dv$ represents the probability that the relative velocity lies between v and $v + dv$. The normalization condition is

$$\int_0^\infty P(v)\,dv = 1$$

1.1.3 Reaction Rate for a Velocity Distribution

For a distribution of velocities, the reaction rate becomes

$$r_{12} = N_1 N_2 \int_0^\infty v P(v) \sigma(v)\, dv$$

The integral is defined as

$$\boxed{\langle \sigma v \rangle_{12} = \int_0^\infty v P(v) \sigma(v)\, dv}$$

and is called the reaction rate per particle pair. Hence,

$$\boxed{r_{12} = N_1 N_2 \langle \sigma v \rangle_{12}}$$

For identical particles,

$$\boxed{r_{12} = \frac{N_1 N_2 \langle \sigma v \rangle_{12}}{1 + \delta_{12}}}$$

where δ_{12} is the Kronecker delta.

1.2 Charaged Particle Induced Reactions

In stars, nuclear charge particle induce reactions are initiated by thermal motion. As gravitational contraction increases the temperature, nuclei acquire kinetic energy and collide with one another. Since these reactions originate from thermal motion, they are called thermonuclear reactions

1.3 Maxwell–Boltzmann Distribution

In most stellar environments the velocity distribution is described by the Maxwell Boltzmann distribution

$$P(v)\, dv = 4\pi v^2 \left(\frac{m_{12}}{2\pi kT} \right)^{3/2} e^{-m_{12} v^2/(2kT)}\, dv$$

The reduced mass is $m_{12} = \frac{m_1 m_2}{m_1 + m_2}$. The kinetic energy is $E = \frac{m_{12} v^2}{2}$. Differentiating

$$\frac{dE}{dv} = m_{12} v$$

The energy distribution becomes

$$P(E)\, dE = \frac{2}{\sqrt{\pi}} \frac{\sqrt{E}}{(kT)^{3/2}} e^{-E/kT}\, dE$$

The most probable velocity is

$$v_T = \sqrt{\frac{2kT}{m_{12}}}$$

and the corresponding energy is

$$E_T = \frac{kT}{2}$$

1.3.1 Reaction Rate Per Charge Particle Pair

Using the Maxwell–Boltzmann distribution, the reaction rate per particle pair is

$$\boxed{\langle \sigma v \rangle_{12} = \sqrt{\frac{8}{\pi m_{12}}} \frac{1}{(kT)^{3/2}} \int_0^\infty E\sigma(E) e^{-E/kT}\, dE}$$

In practical applications, the quantity tabulated is

$$N_A \langle \sigma v \rangle$$

where N_A is Avogadro's number. The units are

$$\mathrm{cm}^3\,\mathrm{mol}^{-1}\,\mathrm{s}^{-1}$$